



Diabetic Retinopathy Debrecen

Donated on 11/2/2014

This dataset contains features extracted from the Messidor image set to predict whether an image contains signs of diabetic retinopathy or not.

Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification

Feature Type

Integer, Real

Instances

1151

Features

19

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 1 citations

 13435 views

Creators

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DOI

10.24432/C5XP4P

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Dataset Information

What do the instances in this dataset represent?

Medical patients

Additional Information

This dataset contains features extracted from the Messidor image set to predict whether an image contains signs of diabetic retinopathy or not. All features represent either a detected lesion, a descriptive feature of a anatomical part or an image-level descriptor. The underlying method image analysis and ...

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Has Missing Values?
No

This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Introductory Paper



[An ensemble-based system for automatic screening of diabetic retinopathy](#)

By B. Antal, A. Hajdu. 2014
Published in Knowledge-Based Systems

Variables Table



Variable Name	Role	Type	Description	Units	Missing Values
quality	Feature	Binary	The binary result of quality assessment. 0 = bad quality 1 = sufficient quality.		no
pre_screening	Feature	Binary	The binary result of pre-screening, where 1 indicates severe retinal abnormality and 0 its lack.		no
ma1	Feature	Integer	ma1 - ma-6 contain the results of MA detection. Each feature value stand for the number of MAs found at the confidence		no

Variable Name	Role	Type	Description	Units	Missing Values
			levels alpha = 0.5, . . . , 1, respectively.		
ma2	Feature	Integer			no
ma3	Feature	Integer			no
ma4	Feature	Integer			no
ma5	Feature	Integer			no
ma6	Feature	Integer			no
exudate1	Feature	Continuous	exudate1 - exudate8 contain the same information as 2-7) for exudates. However, as exudates are represented by a set of points rather than the number of pixels constructing the lesions, these features are normalized by dividing the number of lesions with the diameter of the ROI to compensate different image sizes.		no
exudate2	Feature	Continuous			no

Variable Name	Role	Type	Description	Units	Missing Values
exudate3	Feature	Continuous			no
exudate3	Feature	Continuous			no
exudate5	Feature	Continuous			no
exudate6	Feature	Continuous			no
exudate7	Feature	Continuous			no
exudate8	Feature	Continuous			no
macula_opticdisc_distance	Feature	Continuous	The euclidean distance of the center of the macula and the center of the optic disc to provide important information regarding the patient's condition. This feature is also normalized with the diameter of the ROI.		no
opticdisc_diameter	Feature	Continuous	The diameter of the optic disc.		no
am_fm_classification	Feature	Binary	The binary result of the AM/FM-based classification.		no

Variable Name	Role	Type	Description	Units	Missing Values
Class	Target	Binary	Class label. 1 = contains signs of DR (Accumulative label for the Messidor classes 1, 2, 3), 0 = no signs of DR.		no

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Additional Variable Information

0) The binary result of quality assessment. 0 = bad quality 1 = sufficient quality.

1) The binary result of pre-screening, where 1 indicates severe retinal abnormality and 0 its lack.

2-7) The results of MA detection. Each feature value stand for the number of MAs found at the confidence levels $\alpha = 0.5, \dots, 1$, respectively.

8-15) contain the same information as 2-7) for exudates. However, as exudates are represented by a set of points rather than the number of pixels constructing the lesions, these features are normalized by dividing the number of lesions with the diameter of the ROI to compensate different image sizes.

16) The euclidean distance of the center of the macula and the center of the optic disc to provide important information regarding the patient's condition. This feature is also normalized with the diameter of the ROI.

17) The diameter of the optic disc.

18) The binary result of the AM/FM-based classification.

19) Class label. 1 = contains signs of DR (Accumulative label for the Messidor classes 1, 2, 3), 0 = no signs of DR.

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Dataset Files ^

File	Size
messidor_features.arff	114.5 KB

Reviews ^

There are no reviews for this dataset yet.

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